EDGE AI: THE BIG IDEA

1. By the end of today I can...

- [ ] explain Edge AI, and how it differs from cloud AI
- [ ] tell fixed-function AI from generative AI
- [ ] name the main parts of a Pi 4 and a Pi 5
- [ ] explain what an operating system does
- [ ] understand router, IP address, host name, `.local`
- [ ] write the OS onto a microSD card with my own settings
- [ ] boot, read the LEDs, log in over SSH, shut down safely

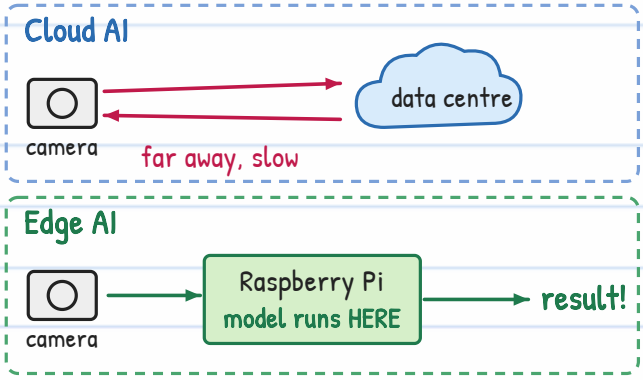
My 2 hours

A big picture	20 min
B the Raspberry Pi	25 min
C OS + network basics	15 min
D prepare the microSD	35 min
E first boot	10 min
F first SSH login	15 min

my notes

2. What is Edge AI?

- **Cloud AI:** my data travels to a data centre, the model computes, the answer travels back.
- **Edge AI:** the AI algorithm runs locally, on the device next to the data: a camera, robot, factory sensor, phone or a Raspberry Pi.
- The "edge" = the edge of the network, where data is born.
- Result: systems that are more **responsive, private and efficient**.



my notes

3. Training and inference

- A **model** is a program whose behaviour is *learned from examples* instead of written by hand.

Stage	What happens	Where in this course
Training	show the model many examples; it adjusts its internal numbers until answers are good. Heavy: lots of data, fast processor.	laptop or cloud (Colab, Edge Impulse)
Inference	give the trained model new data (photo, sound, question); it answers. Light: milliseconds to seconds.	on the Raspberry Pi

Typical project: train (or download) once, shrink the model to fit a small device, then run inference on the Pi again and again.

my notes

WHY THE EDGE? TWO FAMILIES OF AI

4. Five reasons Edge AI matters

Advantage	What it means in practice
Low latency	answer in milliseconds, no network round trip: a robot reacts *now*
Privacy	sensitive images / audio stay on the device
Network independence	keeps working with no Internet: tunnel, field, ship, factory
Less bandwidth	send only the result (a few bytes), not a video stream
Energy efficiency	small optimised models on small hardware use little power

The price of the edge: small processor, limited memory, limited power budget -> models must be **small and optimised** (e.g. lower numerical precision). Fitting a model to the hardware is a theme of the whole course.

my notes

5. Two families of AI

Fixed-function (reactive)

- analyses a specific input using predetermined patterns / rules
- responds only to particular triggers
- consistent, predictable output
- excels at structured tasks: classify, detect, recognise
- works within its boundaries

Generative (proactive)

- creates new content: text, descriptions, solutions
- understands context
- keeps a dialogue going
- adapts to situations beyond its training

	Fixed-function	Generative
Core focus	data analysis, assessment, automation	content creation, anticipation, dialogue
Examples	fraud detection, spam filters, image recognition, counting objects	chat assistants, image generators, predictive maintenance, smart assistants
Typical use	banking, medical diagnostics, security	content, customer support, design, automation
In this course	Phase 2: classification, detection, counting	Phase 3: language models, vision-language, agents

Rule of thumb: fixed-function = efficient, predictable, low resources, well-defined task. Generative = creative, personalised, anticipatory. Both gain from the edge: lower latency, privacy, reliability offline, efficient resources.

my notes

6. What the whole course covers

1. **Essential setup** of the Raspberry Pi (this week)
2. **ML foundations** on Google Colab
3. **Computer vision:** classification, detection, counting
4. **Small language models:** run, optimise, add retrieval (RAG), tools, agents
5. **Vision-language models**
6. **Physical computing:** sensors and actuators through GPIO
7. **Capstone project** and how to teach it

my notes

THE RASPBERRY PI

7. A computer on a single board

- **Single-board computer (SBC):** a complete computer on one circuit board: processor, memory, network + video connectors, pins for electronics.
- Its main chip is a **system on a chip (SoC)**: CPU cores, graphics and controllers in one package.
- No built-in disk. The OS and my files live on a **microSD card**. Swap the card = swap the computer's whole "personality".

my notes

8. Why the Pi suits Edge AI

Feature	What it gives me
Computational power	multi-core ARM CPUs, up to 8 GB RAM on newer models
GPIO, 40 pins	each pin reads a signal (button, sensor) or drives one (LED, relay)
Connectivity	Wi-Fi, Bluetooth, Gigabit Ethernet, USB
I2C, SPI, UART	serial buses for sensors, displays, other chips
Soft real-time	control / signal tasks that tolerate small timing variation
Power efficiency	suits battery-powered and always-on designs
Full Linux + community	standard tools, huge community, low cost

my notes

9. What engineers build with a Pi

- **Embedded systems:** prototyping devices
- **IoT:** sensors using MQTT, CoAP, HTTP(S)
- **Control systems:** feedback loops, PID, actuators
- **Vision + AI:** OpenCV, TensorFlow Lite
- **Data acquisition:** sensor logging, real-time analysis
- **Robotics:** controllers, motion planning
- **Signal processing:** real-time filtering, DSP
- **Network security:** VPNs, firewalls, test labs

my notes

10. Models compared

	Zero 2 W	Pi 4 Model B	Pi 5
CPU	1 GHz, 4 cores, Cortex-A53, 64-bit	1.5 GHz, 4 cores, Cortex-A72	2.4 GHz, 4 cores, Cortex-A76
RAM	512 MB	up to 8 GB	2 / 4 / 8 GB (16 GB later)
Wi-Fi / BT	2.4 GHz only, BT 4.2	dual-band, BT 5.0	dual-band, BT 5.0
Power	5 V / 2.5 A micro-USB, ~0.6 W idle	5 V / 3 A USB-C	5 V / 5 A (27 W) USB-C, ~3-3.5 W idle
Best for	tiny battery projects, small models	vision projects (language models: only tiny ones)	vision + the largest SLMs (8 GB best)

my notes

PI 4 OR PI 5: WHAT CHANGES

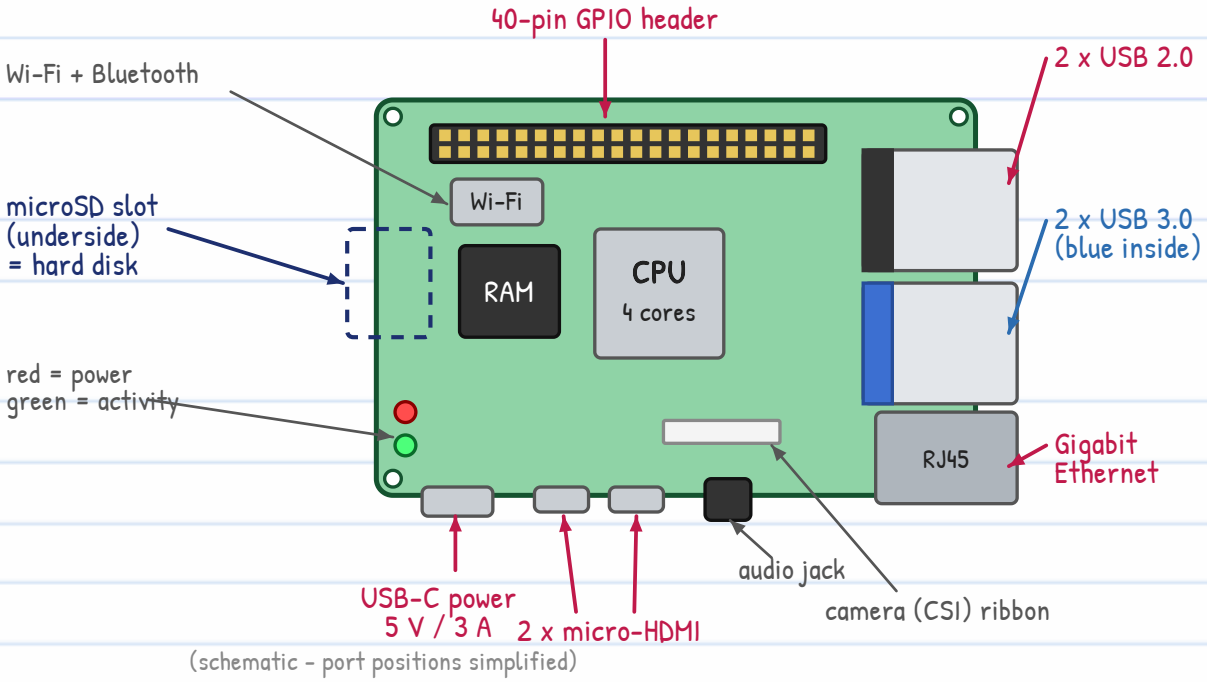
11. In practice

Topic	Pi 4	Pi 5
Speed	Cortex-A72, 1.5 GHz	Cortex-A76, 2.4 GHz: clearly faster
Memory	up to 8 GB	2, 4, 8 GB
Power supply	USB-C 5 V / 3 A	USB-C 5 V / 5 A (27 W); a 3 A supply limits USB devices to 600 mA
Power button	none: plug in = on	yes
Cooling	heatsink / fan case advisable for long runs	hotter; fan connector; active cooler advisable
Camera cable	standard 15-pin ribbon	smaller 22-pin connector: check the cable
Operating system	older and newer releases	needs a recent Raspberry Pi OS
In Imager	Device = Raspberry Pi 4	Device = Raspberry Pi 5
SSH, commands, updates, shutdown	**identical** on both boards	

Which one? The vision phases run well on both boards. A **Pi 4 does not run language models well** (only very small ones, slowly); for Phase 3 use a **Pi 5, 8 GB recommended**. With only a Pi 4, do the whole course and treat Phase 3 as a guided demonstration.

my notes

12. Anatomy of the Raspberry Pi 4

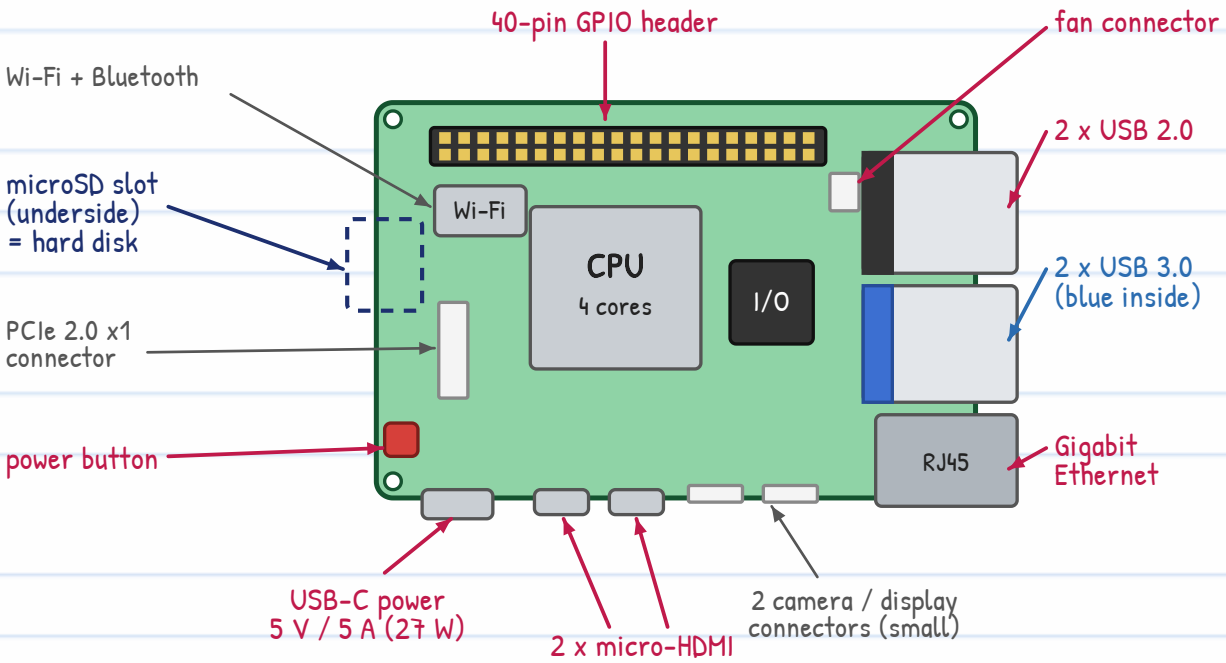


(schematic - port positions simplified)

my notes

MEET THE BOARDS

13. Anatomy of the Raspberry Pi 5



(schematics: port positions are simplified - compare with the real board)

my notes

14. What each part does

Part	What it does	Pi 4 / Pi 5
CPU (SoC)	runs the OS and my code; does the neural-network maths	BCM2711 A72 / BCM2712 A76
RAM	working memory: a model must fit in it	up to 8 GB / 2-8 GB
microSD slot	holds the card (underside)	both
GPIO 40 pins	sensors, LEDs, buttons, relays	both, same layout
USB	keyboards, webcams, mics, storage; blue = USB 3	2 x USB 3 + 2 x USB 2, both
Camera / display	ribbon connectors	1 CSI + 1 DSI / 2 small connectors
micro-HDMI x 2	monitors; use the port nearest USB-C first	both
Power button, PCIe, fan	on / off; fast expansion; cooling fan	Pi 5 only
Status LEDs	red = power, green = activity	both

my notes

my notes

WHAT I NEED ON THE DESK

15. What I need on the desk

Required

- Pi 4 or Pi 5
- microSD card, **32 GB+**, reputable brand, A1 / A2 rating
- correct USB-C supply (Pi 4: 5 V / 3 A; Pi 5: 5 V / 5 A)
- card reader (or laptop slot)
- laptop with Wi-Fi

Optional

- monitor + micro-HDMI cable
- USB keyboard + mouse
- Ethernet cable (fallback)
- heatsink / fan case (active cooler on Pi 5)

Handling: hold the board by its edges; no metal surface while powered; no wet fingers. Cheap or fake cards and weak chargers cause strange failures.

my notes

16. My turn

- [ ] find every part on my real board; take a top-down photo, label it, and paste or sketch it here

my notes

OPERATING SYSTEM BASICS

17. What is an operating system?

- Software that **manages the hardware and the software resources** and gives programs standard services. It sits between the hardware and my applications.

Your programs - Python scripts, Jupyter, AI models

Shell / desktop - where I type commands or click

Kernel + drivers - processes, memory, files, devices

Hardware - CPU, RAM, microSD, Wi-Fi, USB, camera, GPIO

It manages

- **processes:** shares CPU time
- **memory:** gives and frees RAM
- **files:** files and folders
- **devices:** via drivers
- **user interface**

Made of

- **kernel:** the core, controls hardware
- **shell:** where I type commands
- **file system:** how data is stored
- **drivers:** talk to devices

my notes

18. Linux, Debian, Raspberry Pi OS

- **Linux:** family of free, open-source operating systems around the Linux kernel. A **distribution** = kernel + tools + libraries + package manager.
- **Debian:** one of the oldest, most widely used distributions.
- **Raspberry Pi OS** (formerly Raspbian): Debian adapted for the Pi's ARM processors. Lightweight, versatile, open source, GPIO support, regular updates.

Versions and Python: releases follow Debian (e.g. Debian 12 "Bookworm"; newer ones exist). Imager offers the current one; the steps are the same. The **Python version** can differ, and some ML packages are slow to support the newest Python. Day 5 handles it with virtual environments. Record my OS and Python versions today.

my notes

19. Which image do I choose?

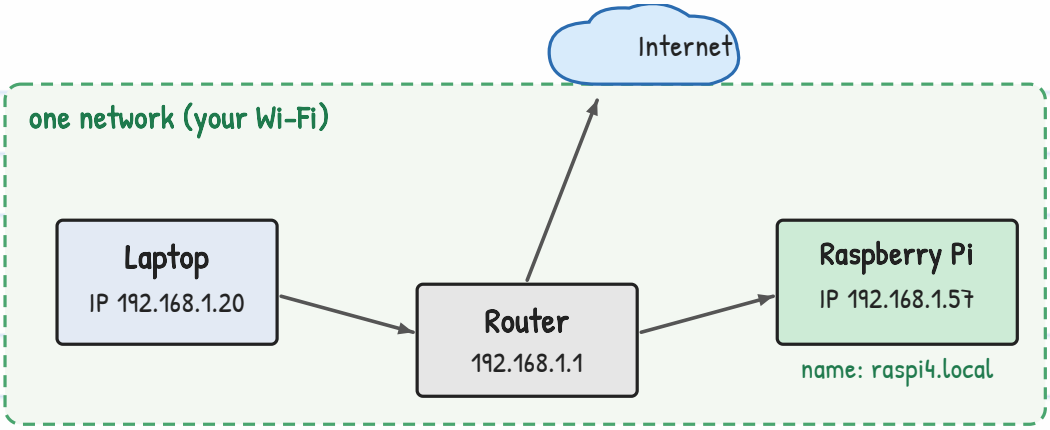
Choice in Imager	What it is	Mine?
Raspberry Pi OS (64-bit)	full system **with desktop**; many AI libraries need 64-bit	YES (Pi 4 and Pi 5)
Raspberry Pi OS Lite (64-bit)	no desktop, saves RAM (for tiny boards like the Zero 2 W)	not needed
32-bit versions	older; fewer AI packages	avoid

64-bit? More memory addressable, full CPU instruction set. Check later with `uname -m`: **aarch64** = 64-bit ARM, `armv7l` = 32-bit.

my notes

JUST ENOUGH NETWORKING

20. How the laptop will find the Pi



IP address
a device's number on the network

Router (DHCP)
hands out addresses automatically

.local name (mDNS)
announced inside the network - works only there

Idea	Explanation
Router	connects my devices (Wi-Fi or cable) to each other and to the Internet
IP address	a device's number on the network, e.g. `192.168.1.57` (four numbers 0-255). `192.168.x.x`, `10.x.x.x`, `172.16-31.x.x` are private: they work only inside my network
DHCP	router service that gives each device an address automatically; it can change from day to day
Host name	friendly name I choose, e.g. `raspi4`
`.local` **name**	devices announce their names inside the network (mDNS), so `raspi4.local` works without knowing the IP. **Only inside one network**; some networks block it
Wi-Fi bands	2.4 GHz: longer range, slower. 5 GHz: faster, shorter range. Pi 4 and Pi 5 use both; Pi Zero 2 W only 2.4 GHz

my notes

21. Networks that make life harder

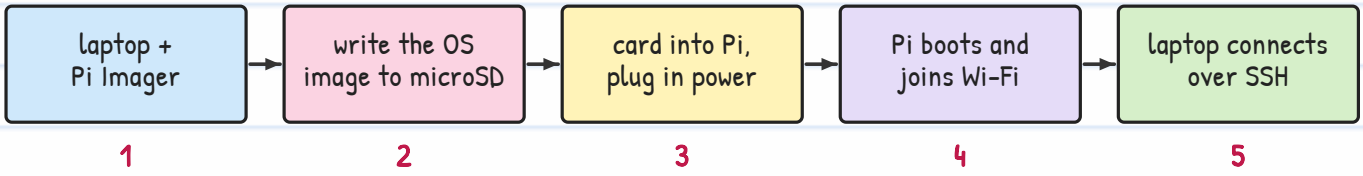
- **Guest networks** and some office / university networks isolate devices: the laptop cannot see the Pi.
- **Enterprise networks** (WPA2-Enterprise) need a user name too: Imager's simple Wi-Fi form cannot do that.
- **Login-page (portal) networks** do not work for a Pi without a screen.
- In all cases: use a **phone hotspot** or an **Ethernet cable** to a home router for the first connection.

my notes

PREPARE THE MICROSD CARD

22. The idea

- At power-up the built-in firmware looks for an OS on the **microSD card**. "Installing the OS" = writing an image (a complete ready-to-run copy of the system) onto the card.
- Tool: **Raspberry Pi Imager** (free; Windows, macOS, Linux). It can also store my **settings** (host name, user, password, Wi-Fi, SSH) so the Pi needs **no monitor or keyboard = headless**.



my notes

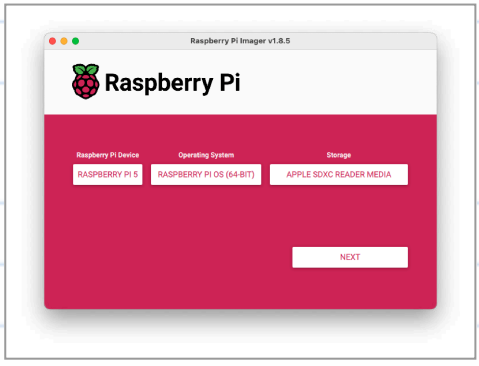
23. Install Imager; before I start

- [ ] download Imager from raspberrypi.com/software, run the installer, open it
- [ ] microSD card (32 GB+) in a reader
- [ ] Wi-Fi name + password ready
- [ ] card backed up: Imager **erases** it
- [ ] other USB drives unplugged (so I cannot pick them by mistake)

my notes

24. Steps on Windows

1. Insert the card. If Windows says "format the disk?" -> **Cancel**.
2. Device = `Raspberry Pi 4` (Pi 5: `Raspberry Pi 5`)
3. OS = `Raspberry Pi OS (64-bit)`
4. Storage = my card (check size + drive letter!)
5. Next -> open the settings (gear icon / "customise")
6. Write -> verify -> "Write successful" -> remove card



Imager with device, OS and storage chosen (example on a Mac, older release)

Careful! Imager erases whatever drive I choose. Wrong drive = lost data.

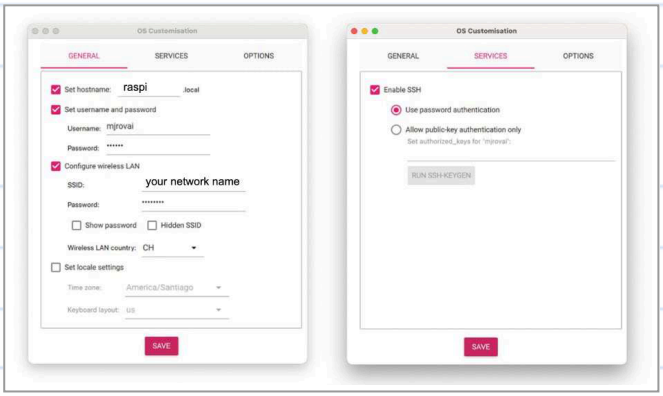
my notes

SETTINGS & WRITING THE CARD

25. The settings, one by one

Setting	What to enter	Why
Host name	`raspi4` (letters, digits, hyphens; no spaces)	the Pi's name; I reach it as `raspi4.local`. A different name for each Pi
User + password	`piuser` + strong password	no default user any more; needed for SSH and `sudo`; case-sensitive; write it on my card
Wi-Fi	SSID + password **exactly** as spelled	the Pi must join by itself at first boot; typos are the #1 failure
Wi-Fi country	`KW`, `EG`, `US` ... (where the Pi runs)	radio regulations differ by country
Zone / keyboard	`Asia/Kuwait`, `us`	correct clock; right characters
SSH	**ENABLE**, password authentication (Services tab)	without it: no remote logins

One Wi-Fi only: Imager stores a single network. A second one (lab and home) is added later from the Pi (Day 2).



General tab (left), Services tab (right)

my notes

26. Write the card

- Confirm the warning: **all data on the card is erased.**
- Imager **writes**, then **verifies**: about 5–15 minutes. Do not remove the card or close Imager.
- "**Write successful**" -> Continue -> remove the card (Imager ejects it).

my notes

FIRST BOOT

27. Steps

1. Card into the slot on the **underside** (contacts toward the board).
2. Optional monitor (micro-HDMI port nearest the USB-C) and keyboard first.
3. Power **last**. **Pi 4:** no power button, plug in = on. **Pi 5:** 27 W supply; it has a power button.
4. Wait a few minutes; it may restart once. **Do not unplug!**

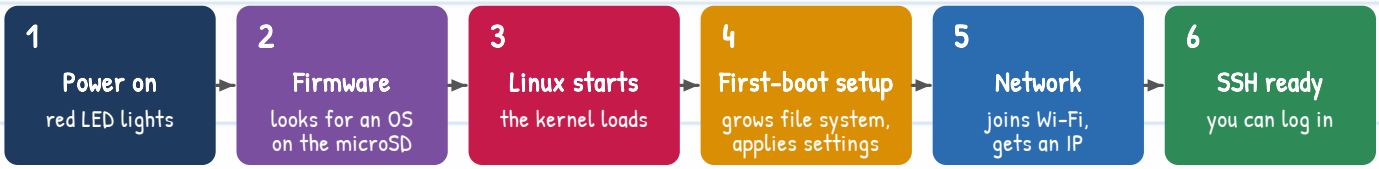
RED steady = power is good

GREEN flickers = Pi reads the card

no red light? check the power supply

my notes

28. What happens during first boot



about 1 to 3 minutes the first time

Step 4 happens only the first time: the Pi grows the file system to fill my card and applies the settings I typed in Imager.

my notes

29. Reading the LEDs

LEDs	Meaning	Do
red steady, green flickering	powered, reading the card: booting	wait 1–3 min
red steady, green never flickered	no OS found	reseat card / re-write / other card
nothing lights up	no power	other cable or charger
random restarts	weak power supply	use the proper charger (Pi 4 5 V/3 A, Pi 5 5 V/5 A)

Heat: idling is cool, but a Pi 5 (and a Pi 4 under load) gets hot in long runs: fit a heatsink or fan (active cooler on a Pi 5).

my notes

FIRST SSH LOGIN

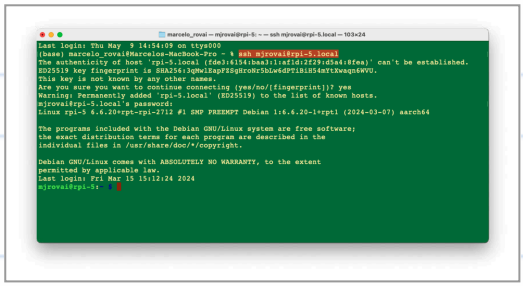
30. What is SSH?

- **SSH** (Secure Shell): the Pi's command line in a window on my laptop. An SSH **server** on the Pi (port 22) and an SSH **client** on the laptop; everything travels **encrypted**. No monitor, keyboard or mouse needed on the Pi.
- Both machines must be on the **same network**.

my notes

31. Steps on Windows

1. Start -> **PowerShell**. Test: `ssh -V`. (Missing? Settings -> Apps -> Optional features -> OpenSSH Client.)
2. Connect (below), by name or IP.
3. First time: type the whole word `yes` to trust this Pi.
4. Password: **nothing shows** as I type: normal!
5. Prompt `piuser@raspi4:~ $` = I am **inside the Pi**.



a first login (example on a Mac)

```
PS C:\> ssh piuser@raspi4.local  (or ssh piuser@192.168.1.57)
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
piuser@raspi4.local's password: (invisible)
piuser@raspi4:~ $ _
```

Mac / Linux: Terminal, same command. PuTTY is an older graphical option on Windows.

my notes

32. Reading the prompt

`piuser` @ `raspi4` : `~` `$`	user @ host-name of the machine I am on (**the Pi**) : current folder (`~` = home) and `$` = ready for a command. `exit` leaves.

my notes

COMMANDS, UPDATES, SHUTDOWN

33. My first commands

Command	What it does	Typical output
`hostname -I`	the Pi's IP address	`192.168.1.57`
`uname -m`	processor architecture	**aarch64** = 64-bit
`cat /etc/os-release`	which OS and release	Debian ... (e.g. 12 bookworm)
`python3 --version`	Python version	Python 3.x.y: write it down
`free -h`	memory	close to 4 GB on a 4 GB board
`vcgencmd measure_temp`	CPU temperature	`temp=46.2'C`

my notes

34. Updating the system

- Software comes in **packages** handled by `apt`. `sudo` = "run as administrator" (asks my password).

```
sudo apt-get update  # refresh the package list
sudo apt upgrade  # install newer versions; answer Y when asked
```

Needs Internet; may take minutes the first time; do it regularly; `sudo reboot` after a kernel update.

my notes

35. Shut down safely, every time

Never just pull the plug. The Pi may still be writing to the card: data loss or a corrupted card.
`sudo shutdown -h now` (halt) `sudo reboot` (restart)
Wait until the **green LED stops flashing and stays dark**, then unplug.

- Start a halted **Pi 4** again: unplug + replug. **Pi 5:** press the power button.
- Files between laptop and Pi (Day 2): USB drive, `scp`, or SFTP (FileZilla / WinSCP).

my notes

WHEN SOMETHING DOES NOT WORK

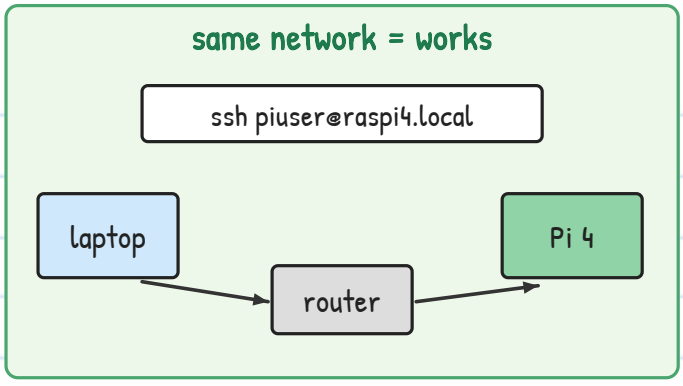
36. Troubleshooting

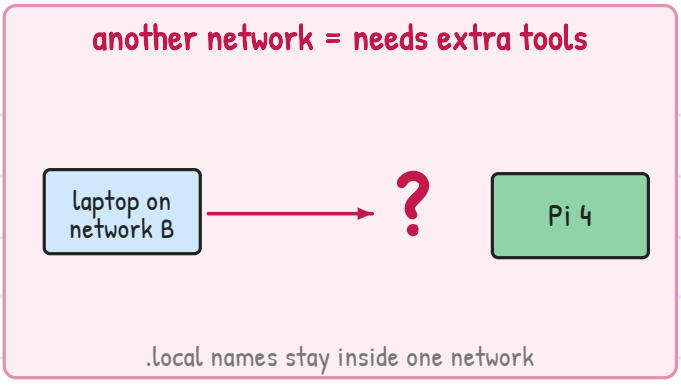
I see	probably	try
`Could not resolve hostname`	Pi not on the network yet; Wi-Fi typo; `.local` blocked	wait 5 min; check Wi-Fi; use the IP from the router's device list; Ethernet / hotspot
`Connection timed out`	different networks; VPN; guest / office Wi-Fi isolates devices	same Wi-Fi; VPN off; hotspot
`Permission denied`	wrong user or password	case matters; retype; or re-write the card
`HOST IDENTIFICATION HAS CHANGED`	I re-wrote the card: the Pi has a new identity	`ssh-keygen -R raspi4.local`
`ssh` not recognized	OpenSSH client missing	Settings -> Apps -> Optional features
red LED, green never flickers	no OS found	reseat; re-write; other card
nothing lights up	no power	other cable / charger
Pi never on the network	Wi-Fi typo, wrong country, hidden / enterprise / portal network	re-write with corrections; Ethernet or hotspot
worked yesterday, not today	router gave a new IP, or other network	use `raspi4.local` or find the new IP

my notes

37. More than one Pi or laptop

- The Pi is the **shared workspace**; each laptop is a **window** into it. Code and notes live on the Pi and in an online repository.
- Learn on one Pi, then **repeat on the other from my notes only**. Different host names (`raspi-lab` , `raspi-home`), same user. One connection card per Pi.





my notes

QUIZ & GLOSSARY

38. Check myself (answer here first!)

1. Cloud AI versus Edge AI: what is the difference?
2. Three advantages of the edge, and one price paid for it.
3. Training versus inference: which one happens on the Pi in this course?
4. Image classification: fixed-function or generative? One example of the other kind.
5. What is a single-board computer? Why does a Pi need a microSD card?
6. Three things an operating system manages.
7. What is an IP address? What is DHCP? Why does `raspi4.local` work only inside one network?
8. Why choose 64-bit Raspberry Pi OS? Which command shows which I have?
9. Why enable SSH in Imager before writing the card?
10. Red LED steady = ? Green LED flickering = ?
11. What can go wrong if I unplug without shutting down? How do I restart a halted Pi 4? A Pi 5?
12. Three differences between the Pi 4 and the Pi 5.

my notes

39. Answers

1 cloud sends data to a data centre; edge computes next to the sensor. 2 latency, privacy, offline operation, bandwidth, energy; price: limited CPU / RAM / power, so small optimised models. 3 training builds the model (laptop/cloud); inference uses it (Pi). 4 fixed-function; generative: language models, image generators. 5 a whole computer on one board; no built-in disk, OS and files on the card. 6 processes, memory, files, devices, user interface. 7 a device's number; router service that hands out addresses; `.local` is announced only inside the local network. 8 many AI packages are 64-bit only; `uname -m` = aarch64. 9 so the Pi accepts remote logins with no monitor. 10 power OK; reading the card. 11 data loss / corrupted card; Pi 4 replug; Pi 5 power button. 12 e.g. speed, power button, power supply, cooling, camera connector, RAM options.

my notes

GLOSSARY, CARD & WRAP-UP

40. Glossary

Term	Meaning
Edge / cloud AI	AI on the device next to the data / in a remote data centre
Training / inference	building a model from examples / using it on new data
SBC, SoC	single-board computer; system on a chip
GPIO, CSI, DSI	general-purpose pins; camera connector; display connector
Image, Imager	a complete OS copy; the tool that writes it to a card
Kernel, shell	the OS core; where I type commands
64-bit, aarch64	full 64-bit processor use; 64-bit ARM
Headless	no monitor, keyboard or mouse on the computer
Router, IP, DHCP	connects a network; a device's number; assigns numbers
Host name, mDNS	a device's name; how names are announced locally (`.local`)
SSID	a Wi-Fi network's name
SSH, terminal	secure remote command line; text window for commands
`sudo` , `apt`	run as administrator; the package manager
Firmware	built-in software that starts the boot

my notes

41. Connection card (one per Pi)

board (Pi 4 / Pi 5, RAM)	
host name / user name	
password hint (not the password!)	
Wi-Fi 1 / Wi-Fi 2	
IP address (it can change)	
OS release / Python version	
uname -m / microSD size	

my notes

42. Day 1 is done when...

- [ ] the Pi boots: red LED steady, green flickers
- [ ] PowerShell shows `piuser@raspi4:~ $`
- [ ] all commands ran and I wrote down OS + Python
- [ ] I shut down with `sudo shutdown -h now`
- [ ] the connection card is filled in

my notes

INSTRUCTOR NOTES

43. A 45-minute class version

Before class: each pair gets a Pi, a card, a reader, a power supply, and a hotspot or router; Imager pre-installed.
10 min big picture (cloud vs edge, two families). **15 min** flash the cards live. **10 min** boot + LEDs. **10 min** SSH + first commands.

- **Typical mistakes:** wrong storage device; Wi-Fi typo; SSH not enabled; guest / enterprise network; unplugging without shutdown; "host identification changed" after re-flashing.
- **Mini-assessment:** a screenshot with `hostname -I`, `uname -m`, `free -h` (Wi-Fi details hidden).
- **Mixed boards:** use the Pi 4 / Pi 5 table; let students swap boards for the second pass.

my notes

44. My own notes, ideas, questions

my notes

Tomorrow, Day 2: remote access like a pro (files, remote desktop, more than one network).

ACKNOWLEDGMENTS & REFERENCES

This course is a compilation: it combines material from the sources below with new explanations, examples, diagrams, exercises and comments written for this course. The original authors are gratefully acknowledged.

[1] Rovai, M. (2025). *Edge AI Engineering: Hands-on with the Raspberry Pi.* Open textbook for the course IESTI05, Federal University of Itajubá (UNIFEI), Brazil. Version of 23 October 2025. Online edition: https://mjrovai.github.io/EdgeML_Made_Ease_ebook/

[2] Rovai, M. (2026). *IESTI05 - Edge AI: Edge Machine Learning Systems Engineering.* Course slides, notebooks and video lectures (second semester 2026), UNIFEI, Brazil. Course videos listed by the author: Class 1 (course introduction) https://youtu.be/QAb3MFHwe78 ; Class 2 (Introduction to Linux) https://youtu.be/QkIHILtl0h4 ; Class 3 (Raspberry Pi setup walkthrough) https://youtu.be/YtjsM36LzVE (lectures in Portuguese).

[3] Rovai, M. (2026). *IESTI01 - TinyML: Machine Learning for Embedding Devices.* Course materials (first semester 2026), UNIFEI, Brazil. https://github.com/Mjrovai/UNIFEI-IESTI01-TinyML

[4] Rovai, M. *TinyML4D:* lectures, books, papers and tutorials for embedded machine-learning education. https://github.com/Mjrovai/TinyML4D

[5] Raspberry Pi Imager: download page and software. https://www.raspberrypi.com/software/

[6] Raspberry Pi Ltd. *Raspberry Pi documentation* (hardware specifications of the Raspberry Pi 4 and Raspberry Pi 5, operating system, remote access). https://www.raspberrypi.com/documentation/

The screenshots of Raspberry Pi Imager and of the first SSH login are adapted from reference [1]. All other figures and the animation were created for this course.

my notes